DSA-110 real-time FRB search at twice the time resolution

Development report, 2026-11-09. The 2x search has been the production search since 2026-11-09 15:13 UTC (op-point 2x-gridded-ns4-cint4, DM plan 0xae1599d449a35b00).

Summary

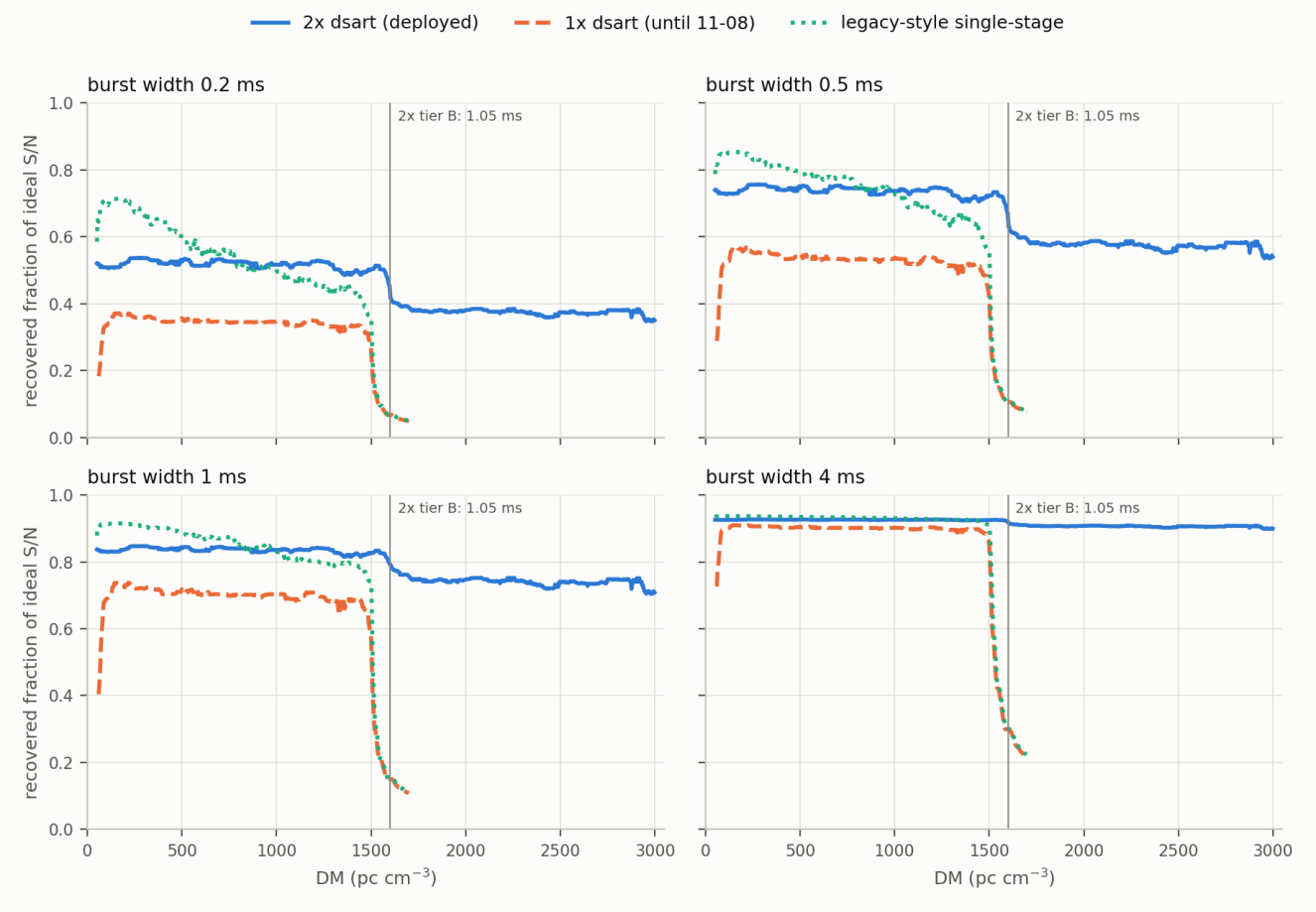
- **What changed.** The search now runs at **0.52 ms** resolution for DM 50-1600 and 1.05 ms for DM 1600-3000. The old (1x) search ran at 1.05 ms for DM 100-1500. Dedispersion now uses 61 kHz channels instead of 244 kHz, and the DM range is twice as wide.
- **What it buys.** In the loss-chain model, a 1 ms burst recovers **0.83** of the ideal matched-filter S/N at DM 100-1500, against 0.70 for 1x (**x1.19**). Narrower bursts gain more (x1.5 at 0.13 ms). At DM 1500 the gain is x1.7, because 1x ran out of trials there. DM 1500-3000 was not searched at all before.
- **Measured.** a real FRB (250924mptq) replayed from recorded voltages lands on the same pixel and DM trial as in 1x, at S/N x1.84. The injection campaign (72 buckets, 0.13-16.8 ms, DM 150-2900) follows the model's recovered fraction with one constant (94% within 20%). Injections now hit their target S/N in normal running; an evening S/N inflation is under investigation (incidents).
- **Real time, with margin.** The correlator path now uses about half its 134 ms block budget. 1x used about 97% of it. Each GPU half of the search is busy 55-59% of the time against a 75% gate, and every search node carries about 11 Gb/s of corner-turn traffic.
- **Status.** Observing in 2x at DEC +16.27 (el 69.04). Cutouts were fixed and redeployed at 19:53 UTC. C3's guard against empty cutouts waits for a 'git pull' on h23 and a dsart_c3 restart. Read the incidents section: 28 real events lost their voltages to C3 on empty cutouts. The first calibrator transit on the new data is 2253+161 at ~03:26 UTC on 11-10; the pipeline should then produce new beamformer weights for you to apply.

1x and 2x side by side

	1x dsart (until 2026-11-08)	2x dsart (production)
Search sampling	1.05 ms	0.52 ms (DM 50-1600), 1.05 ms (1600-3000)
Dedispersion channels	244 kHz (8 native channels summed)	61 kHz (2 summed); stage 1 at 0.26 ms
DM range / trials	103-1496, 8 coarse x 25 fine = 200	50-2996, 24 coarse, 758 fine (two tiers)
Corner-turn format	cint8, per-row max-abs scale (lossy above ~50 sigma)	4-bit gridded cells + 8-bit heavy cells, robust scale, sqrt(n) weighting
Network	~5 Gb/s per search node, 2-fragment frames	~11 Gb/s per search node, single-datagram frames, paced C sender
Search ownership	8 halves, contiguous DM ranges	8 halves, interleaved DMs (3 coarse DMs each)
Raw data kept for cutouts	15 GB ring + cube files	20 s of every DM (29 GB per node): cutouts on demand
Correlator block time	~129 ms of 134 ms (97%)	p50 61-72 ms, p99 67-86 ms of 134 ms (45-64%)

Sensitivity vs DM

The figure compares three searches using one loss model: dedispersion trial spacing, sampling, and intra-channel smearing, as in the time-resolution study. The 2x curve includes the deployed corner-turn format's 0.55% quantisation cost. The legacy-style curve is a single-stage search at 0.26 ms with 244 kHz channels, which is what the legacy dsaX beam search did.



Recovered fraction of the ideal matched-filter S/N vs DM at four burst widths (running median over DM). Blue: 2x (deployed). Orange dashed: 1x. Green dotted: legacy-style single-stage search. The vertical line marks the 2x tier boundary at DM 1600; above it 2x samples at 1.05 ms.

How to read it

- **At 1 ms and wider, 2x is the best of the three at every DM.** At low DM it matches the legacy-style search to within a few percent. Above DM ~800 it beats it, because two-stage dedispersion with 61 kHz channels smears less than 244 kHz single-stage.
- **Below 0.5 ms, at low DM, the legacy-style search is still ahead.** At 0.2 ms it recovers 0.7 against 0.52 for 2x. Its 0.26 ms sampling beats 2x's 0.52 ms. Matching it needs the 4x option, which is network-bound (see compromises). 2x still recovers 1.5x more than 1x there.
- **DM 1500-3000 is new sky.** 1x and the legacy-style search fall off at DM 1500. 2x keeps 0.72-0.75 at 1 ms up to DM 3000.

DM band	2x 0.5 ms	1x 0.5 ms	legacy 0.5 ms	2x 1 ms	1x 1 ms	legacy 1 ms	2x 4 ms	1x 4 ms
50-100	0.74	0.25	0.53	0.84	0.35	0.63	0.93	0.62
100-500	0.74	0.55	0.82	0.84	0.72	0.90	0.93	0.90
500-1000	0.74	0.53	0.76	0.83	0.70	0.86	0.93	0.90
1000-1500	0.73	0.51	0.66	0.83	0.68	0.80	0.93	0.89

DM band	2x 0.5 ms	1x 0.5 ms	legacy 0.5 ms	2x 1 ms	1x 1 ms	legacy 1 ms	2x 4 ms	1x 4 ms
1500-2000	0.61	0.13	0.14	0.76	0.18	0.20	0.91	0.35
2000-3000	0.57	-	-	0.73	-	-	0.90	-

Mean recovered fraction per DM band (model). A dash marks a band that search does not cover.

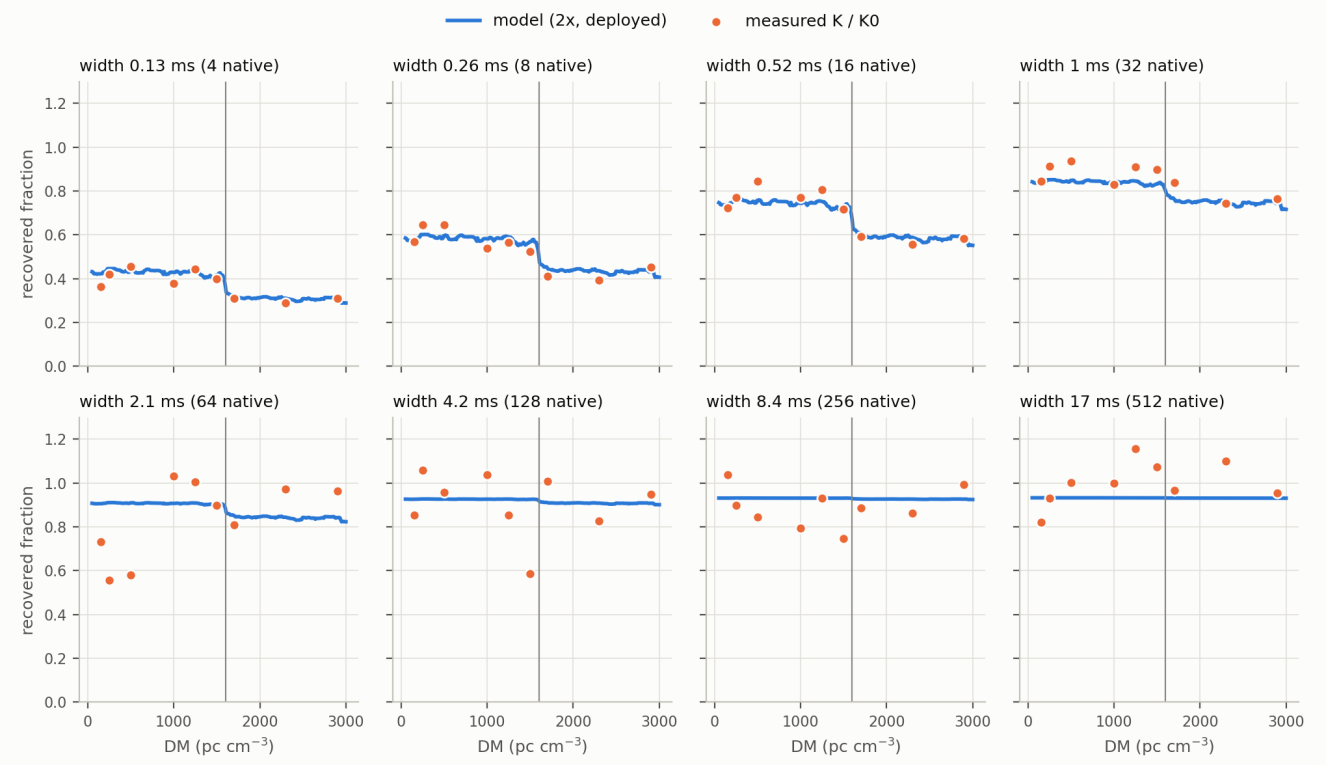
What was measured

A real FRB through both searches

FRB 250924mptq (DM 404.7) was run from recorded voltages through both the old and the new searches. 2x put it at exactly the same pixel as 1x, at the predicted DM trial, with **S/N 266 against 145 (x1.84)**. This is the model's x1.25 at 1 ms, multiplied by the removal of 1x's lossy wire compression for bright bursts (0.68 at ~150 sigma per sample).

Injection calibration campaign

72 (DM, width) buckets were probed: DM 150-2900 across both tiers x 8 widths, 0.13-16.8 ms (4-512 native samples), 15:37-18:40 UTC on 11-09, each probe at the same fluence. The calibration factor K = S/N x sqrt(width) / fluence is proportional to the fraction of ideal S/N the search recovers. So, scaled by **one** constant, the measured K should follow the model's curves; it is a direct test of the whole loss chain. **It does:** 94% of the 72 buckets lie within 20% of the model (16-84% range 0.91-1.11), across 0.13-16.8 ms and DM 150-2900. That includes the drop at the tier boundary (DM 1600) for bursts shorter than 1 ms. The 16.8 ms buckets and the late 8.4 ms ones were first probed at 18:00-18:40, as the evening S/N inflation set in (incidents), and read ~2x high. They were re-measured after the 19:53 restart, as were five buckets whose probe took K from an early, weaker match. After the restart, a check at DM 500 (fixed position, two shots per point) read S/N within 10% of the K prediction at 0.13, 0.52 and 4.2 ms (S/N 18-25); bursts below S/N ~9 were not detected, as expected. A validation grid on the re-measured buckets, at target S/N 20, read 14-27 (one probe per bucket gives K to roughly +-25%). The grid phase that followed (one shot per bucket at a target S/N 20, 18:40-19:52) came out 1.4-14x too bright. It was not the calibration: the same requests after the 19:53 restart gave S/N 19-21 for target 20 (incidents: S/N inflation). The calibration (K per bucket) is stored for the injection bot.

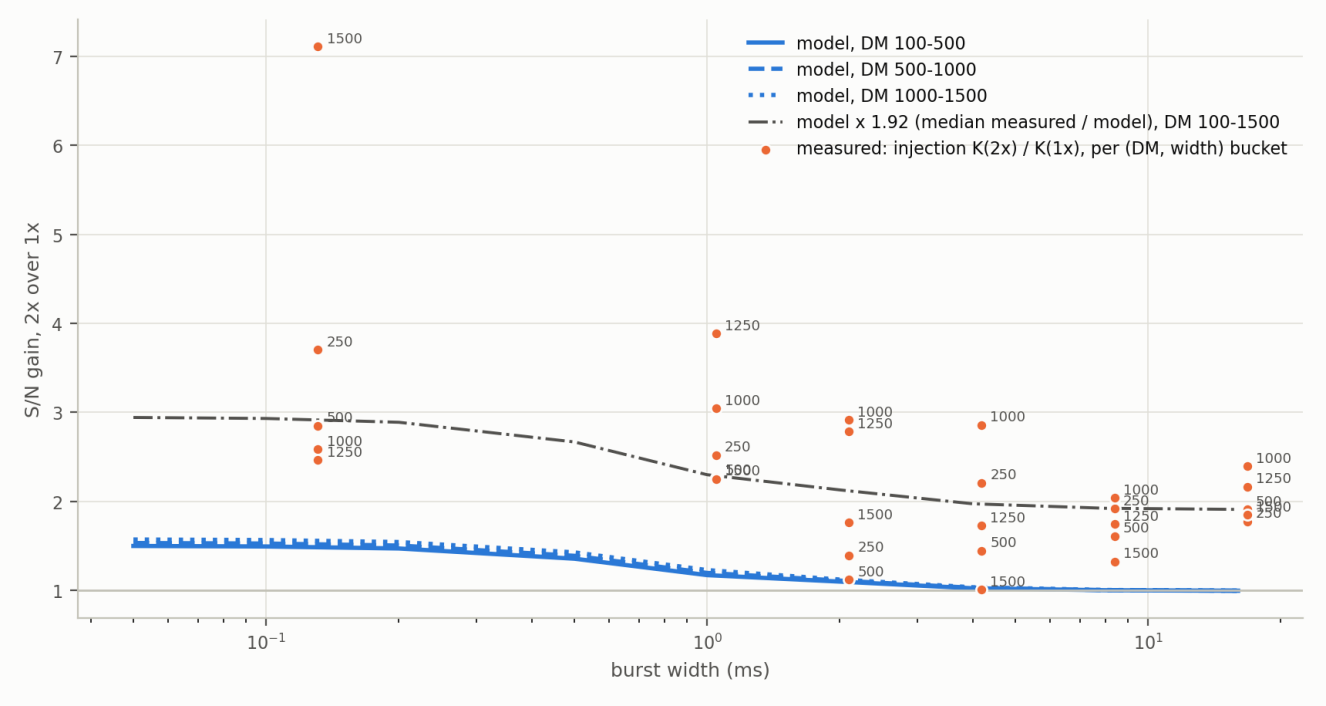


Measured injection K per (DM, width) bucket divided by one fitted constant K0 (orange), against the model's recovered fraction of ideal S/N for the deployed 2x search (blue), one panel per injected burst width (Gaussian FWHM). The vertical line is the tier boundary at DM 1600.

Against 1x

The same fit for the 1x calibrations (30 buckets measured on several days before 11-08, against the 1x model) gives 2x / 1x = **~1.83** (1.8-1.9) in S/N per unit injected fluence, on top of the model's ratio. The 1x calibrations scatter more around their model (63% within 20%), and they come from other days, pointings and gains, so this factor is indicative only. It has the same size as the one same-data comparison, FRB 250924mptq (x1.84 measured against x1.25

modelled). Both say 2x recovers more, relative to 1x, than the model's dedispersion and sampling terms alone predict. The replay attributes the extra to 1x's lossy 8-bit corner turn for bright bursts.



Model S/N gain of 2x over 1x vs burst width for three DM bands (blue). Orange: the ratio of the 2x K to the last 1x K of the same (DM, width) bucket, labelled by DM. The 1x values were measured on other days (gains, pointing, flagging differ); read the spread, not individual points.

Real time in production

Snapshot of the x2 health page in production, 2026-11-09 ~17:15 UTC, two hours after the start. Every correlator node and every search half is inside its real-time gate with wide margin. The only red entry is RX loss on n01/n02 (4-7e-4), which is the known fault on 2-3 fabric paths into those nodes. It was present before 2x, and fan-in absorbs it.

Quantity	Gate	Production (16 corr nodes, 8 halves)
Correlator block time, p50	mean <= 85% of 134.2 ms	61-72 ms (45-54%)
Correlator block time, p99	<= 134.2 ms	67-86 ms
Corner-turn TX per corr node	-	2.70-2.82 Gb/s
RX per search node	<= 80% of clean capacity (17 Gb/s)	11.0-11.6 Gb/s
RX loss per search node	<= 1e-4 (or the 1x value)	n09, n13: 0; n01: 6.9e-4; n02: 4.2e-4
Search GPU duty per half	<= 0.75 (aim 0.6)	0.55-0.59
Cubes per second	nominal 14.90	99.94-100% of nominal

What made it possible

A twice-as-fine search is about 4x the work per second in every stage: twice the samples, four times the DM trials, and a corner turn carrying twice the data. Every stage had to get faster, not just the search.

Correlator (corr_fast, one GPU per node)

- **A new GEMM** computes only the core baselines the search images. It loads and calibrates voltages directly in the multiply, and writes fp16 output in time-minor order. This replaced separate cal, mask and copy kernels.
- **A new stage-1 dedispersion kernel** runs at 0.26 ms. It is driven by a cell lookup table, stages each data window in shared memory once, and computes every DM of a tier in registers. Using four sub-bands (n_sub 4) halves its cost for under 0.5% of S/N.
- **A GPU packer** codes each value in 4 bits against a robust running noise scale, weighted by sqrt(n) so every value carries equal noise. The heaviest cells, about 1% of values but 20% of the weight, travel at 8 bits so bright bursts never clip.
- **A C sendmmsg sender** runs outside Python's lock, paced in 16 KB bursts. With 512 KB bursts, the incast microbursts overflowed the search NICs' port buffers and lost 1-5.6% of the data; 16 KB pacing gives zero NIC discards up to 17 Gb/s per search node.
- **Asynchronous host-to-device copies, deferred host reads and pre-allocation** take the per-block GPU time from about 129 ms to about 50 ms.

Search (8 GPU halves on 4 nodes)

- **A per-DM raw ring** holds 20 s of every owned DM. It replaces the 1x cube files, and cutouts for C2 events are re-imaged on demand from it.
- **The owner engine** is a chain of custom kernels: unpack, combine through a lookup table, a fused FFT imager (column then row inverse FFT, mask and scale in one pass), and a register-tiled detector with a B-spline filter bank of 8 widths (0.52-33 ms). It runs at 3.5-3.6 us per 256x256 image, all-in.
- **Interleaved DM ownership:** half h owns coarse DMs d with d mod 8 = h. Every half sees both tiers, so load is balanced.
- **Fan-in on data time, not wall-clock time,** releases a cycle when 60 of the 64 streams are complete. The known fabric loss on n01/n02 therefore never stalls the search.
- **Priming:** each engine runs two throwaway cycles on in-memory data before attaching. This fixed a start-up lag-recovery seek found in the first live fleet run.

Downstream, monitoring and operations

- C1, C2 and C3 work in external-clock units (0.524 ms samples). C2 never discards an event with missing cubes, and cutouts come from the raw ring.
- Injections use a calibration factor K per (DM, width) bucket, so the bot can inject at a chosen S/N across the whole width range.
- One code base supports both op-points; configs are pushed per op-point to etcd. The web UI Start offers only 2x, and an x2 health page shows block times, duty, loss and fan-in.
- A test-capture tool replaces the Grafana drop-counter check after SNAP restarts. It also measures each SNAP's sequence offset directly.

Compromises

Everything below was either decided with you at a checkpoint or found during deployment. Costs are relative to an ideal implementation of the same design.

1. **Corner-turn format: 4-bit gridded cells with four sub-bands (gridded ns4 cint4), about 0.6% of S/N.** You chose it on 2026-10-29 from the measured network capacity. It recovers 0.994 of the dedispersion model and needs 14 Gb/s of clean ingress per search node, the most network margin of the candidates. The heaviest cells travel at 8 bits so bright bursts do not clip. Even so, the 2x wire costs less S/N than the 1x wire, which lost 1.2%.
2. **Four sub-bands in stage 1 (n_sub 4): under 0.5% of S/N on average, up to 1.7% for narrow bursts at DM 100-500.** This halves the stage-1 kernel's cost. It was the plan's first fallback for the correlator budget.
3. **0.52 ms, not 0.26 ms.** 4x resolution is network-bound: corner-turn traffic grows as 1/T^2, and the fabric cannot carry it without the on-board 10G ports. As a result the legacy-style single-stage search is still ahead below 0.5 ms at low DM (0.70 vs 0.52 of ideal at 0.2 ms).
4. **DM 1600-3000 is searched at 1.05 ms (tier B).** Bursts narrower than ~1 ms lose ~25% there, as the injections confirm. Searching it at 0.52 ms would double tier B's cost. At those DMs intra-channel smearing and scattering usually broaden bursts past 1 ms anyway.
5. **The detector amplitude regime is carried over from 1x unchanged.** That means input clip 250 sigma, sigma floors, and the injection saturation guard at 240. S/N is linear only up to ~35 sigma and rails at ~250 x sqrt(width). The 2026-07-22 attempt to change it produced an all-night RFI storm, so linearity was left out of scope by decision.
6. **Trial spacing tolerance 1.25.** Midway between two DM trials a narrow burst loses up to 1/sqrt(1.25), about 11%; the average loss over DM is a few percent. This is the plan's op-point and is included in the model curves.
7. **A 20 s raw ring instead of cube files.** Cutouts must be copied out of the ring within ~11 s of C2's request, and the producer now copies at accept time. A window that has already rotated out is reported TOO_LATE rather than archived.
8. **The fabric's packet loss into n01/n02 is absorbed, not fixed.** 2-3 of the 64 paths into those nodes lose ~3e-3 (node totals 4-7e-4) because of a faulty inter-switch link. Data-time fan-in (60 of 64 streams) means it never stalls the search; the S/N cost is ~0.05%.
9. **The web-UI arm epoch is rounded down to the 32-sample grid by corr_fast**, and the search mirrors the rounding. Labels stay exact. Rounding `compute_arm_seq` itself would change 1x too, so it was left alone.
10. **Plan steps not done as written.** The 1x/2x web-UI drill was dropped, because you asked for 2x to be the only option. The >= 12 h soak is the production run itself, since 15:13 UTC on 11-09. The science validation grid was replaced by the injection calibration campaign, which covers more DMs and all eight filter widths.

Found and fixed while deploying

- **Voltage dumps wrote 0 blocks.** 2x candidate times are absolute SNAP-epoch sample numbers, but voltage_retention expected capture-relative ones. It now converts them (--event-specnum-epoch snap); ring block 0 is the first armed page.
- **The Sky-tab export crashed** on the 2x sky object (an attribute only 1x has). Fixed by duck-typing.
- **The etcd read-back check failed** on configs with integer keys. The canonical hash now normalises through a JSON round trip.
- **The restart helper armed too early.** It matched old 'ready' lines in appended child logs. It now arms only after every ready sentinel is newer than the Start.
- **Search start-up seek.** The first live fleet run lagged at attach. The engines now run two throwaway cycles on in-memory data first (priming).
- **The injection campaign's probe saturated** at DM 150, width 4 (S/N 516 > rail). The probe is now repeated once at a fluence scaled to S/N ~40.

Incidents on the first day of 2x observing

- **28 real events lost their voltage dumps (11-09 15:49-19:52 UTC).** Every cutout after the first arrived empty (next item), and C3 (flag_only off) applied its rules to the all-zero planes, firing no-peak, image-offset, DM-edge and cube-unconfirmed. It REJECTed 28 real events, broadcast deletion of their staged voltages and deleted their raw windows. Most were the low-DM storm pattern (DM 120-180, S/N 11-25), but not all: 261109boeh (DM 753), grgx (DM 1119), jbbq (S/N 19.4, DM 386) and ayay (DM 1524). The voltages cannot be recovered. Fixed: C3 now fails open (KEEP) when the trigger cutout holds no valid rows (commit 8c5087a; needs 'git pull' on h23 and a dsart_c3 restart). Six events dumped just before the fix still had empty cutouts and were waiting for C3; their cutouts were moved aside (cutouts_lapped_empty), so C3 keeps them.
- **Every cutout from 15:36 to 19:54 was empty.** Each request took ~146 s and requests arrived faster, so every search half stayed backlogged. A request's rows were read only when it was served, by which time the 20 s raw ring had overwritten them. The reader marks such rows invalid and nothing errors. The ~9 GB raw window archived per event in that period holds invalid rows only. Fixed and deployed at 19:53 (6209579, 7e2f11b). First event after: cutouts in 27-32 s, all rows valid.
- **S/N inflation in the evening.** Injections at a fixed fluence read normal S/N from 15:30 to 17:30 and after the 19:53 restart. From about 18:00 to 19:52 they read 1.4-14x high, depending on width: for example a 0.52 ms burst at DM 1000 read S/N 84 against 21 after the restart. That is about 3 h after the start. Real events in that window were judged on inflated S/N too. The cause is not yet known; a slow drift in the detector's noise normalisation is the leading suspect. Every half's per-DM sigma is being logged each minute overnight (x2_sigma_log_20261109.jsonl), with a fixed reference injection every 30 min (x2_reference_shots_20261109.jsonl), both in /dataz/dsa110/operations/inject/.

Operating it

- **Start / stop:** web UI Control tab. Start offers only 2x (2x-gridded-ns4-cint4). Use 're-apply op-point' after any config push. Arm only after every corr node has written a fresh ready sentinel (/tmp/dsart-corr-fast-*.ready, a few minutes after Start). The child logs are appended, so an old 'ready' line there does not count.
- **Updating node code:** Control tab, update_dsart, **branch x2/main, force on**. Never leave the branch blank: blank resets the nodes to origin/main, which is the 1x code. The h23 live checkout is on branch x2-live tracking origin/x2/main; 'git pull' there updates C2/C3/dashboard code (restart the unit or the dashboard afterwards).
- **Changing configs:** edit configs/opoints/2x/*.yaml, then run tools/ops/push_dsart_to_etcd.py --opoint 2x-gridded-ns4-cint4 --plan /home/ubuntu/data/dm_plans/dm_plan_tiered_ns4.npz --write. C2/C3/the inject bot read the yaml from the h23 checkout, so 'git pull' there and restart them.
- **After a SNAP restart:** tools/x2/snap_capture_check.py replaces the Grafana drop-counter check. It captures packets on every corr node and reports per-SNAP sequence offsets and drop rates.
- **Health:** the dashboard's x2 health page (/api/x2/health) shows correlator block time, per-half GPU duty, cube rate, per-node RX loss and cutout latency against the real-time gates. n01/n02 show a loss 'FAIL' at 4-7e-4; that is the known fabric fault, not the search.
- **Injections:** the bot injects at a requested S/N using per-(DM, width) calibration K from this campaign. Widths 4-512 native samples (0.13-16.8 ms), DMs 150-2900. Re-run tools/x2/inject_campaign.py --phase calibrate after any change to gains, beamformer weights or the detector.
- **New beamformer weights:** as you planned, restart with the new weights once a good calibration is in (Stop, update weights, Start with re-apply, arm).

Open items and recommendations

1. **Find the S/N inflation.** Look in the overnight sigma log and the reference shots for the onset, and check the Layer-1 / Layer-2 noise estimates against the 2026-09-23 sigma-feedback flood class. Until it is understood, a restart resets it. The search halves' sigma history is not in Influx (the 1x search_rt_noise measurement has no 2x writer); add it to the pusher.
2. **Pull C3's fail-open guard on h23** (git pull in the live checkout, then systemctl --user restart dsart_c3). Until then C3 still REJECTs on an empty cutout; empty cutouts should no longer occur after the 19:53 fix.
3. **C2 needs a 2x retune.** In the first 2.2 h of 2x, C2 dumped 32 real events (~15 per hour; 1x: 3-4.5 per hour). 23 of them were at DM 100-200 with S/N 11-15, i.e. faint, low-DM and impulsive, the RFI signature. Another 126 were suppressed by the rate limiter. 2x searches ~4x the trials at twice the rate with 1x's 11-sigma narrow trigger. Proposed: raise the bar for DM < 300 at narrow widths, checking the change against the injection completeness measured here.
4. **Raw-window retention needs your decision.** Each real C2 event archives ~9 GB of raw corner-turn data (all DMs, +-0.5 s) under /dataz/dsa110/candidates/<event>/raw. Nothing prunes it, and /dataz has ~2.3 TB free. Options: keep N days, keep only C3-KEEP events, or archive only for bright classes. Raw windows written 11-09 15:36-19:54 hold only invalid (lapped) rows and could be deleted.
5. **Fabric fault into n01/n02:** check the Arista inter-switch/LAG member error counters. 2-3 paths per node lose ~3e-3 because of CRC errors, and the faulty paths change with flow hashing.
6. **Search nodes' antennas.out is not refreshed by calibration pushes.** If a calibration ever changes the E/N header, every 2x stream arrives PATTERN_MISMATCH and the search goes blind (fail-loud).
7. **Verify C3 on 2x cutouts** now that they arrive in time, and check its R-rule thresholds in ms / DM density against a few days of real events.
8. **4x resolution** needs the corr nodes' on-board eno2 10G ports as a second corner-turn path (a slot move does not help).
9. **Merge x2/main into main** after your review. The nodes and h23 run x2/main today.

10. **Voltage dumps collide.** The corr-node writer is serial and the voltage ring holds ~25 s, so the second of two dumps a few seconds apart loses its window (261109svej: 0 bytes written; the first dump took ~27 s). At 2x's dump rate this happens in every storm. Write dumps concurrently, or copy the window first and write it afterwards, as the cutouts now do.
11. **The calibration endpoint takes K from the first injection match.** For 5 of 73 probes a weaker early match set K 2.6-3.7x low (re-measured). Have it wait for the settled best match.
12. **C3 throughput.** C3 handles one event at a time and a KEEP waits for voltage collection and a filterbank (~10-20 min), so it manages a few events per hour, while 2x dumps ~15 real events per hour plus injections. On 11-09 evening it had a 10-event backlog. Retuning C2 (above) helps; also consider running voltage collection and filterbanks off the decision path, and taking events oldest first rather than alphabetically.

Sources: docs/x2/ (SPEC.md, signoff_ledger.md, g1_results.md, g2_results.md), docs/timeres_study/snr2x (model), docs/x2/report/data (campaign jsonl, 1x K snapshot), /dataz/dsa110/operations/inject (campaign, linearity sweep, overnight sigma log and reference shots), x2 health page snapshots.